# Data Science and Artificial Intelligence

# Machine Learning

## Nearest Neighbour

Lecture No. 1

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# Recap of Previous Lecture

- Topic — Recall
- Topic — Precision
- Topic — f1 score
- Topic — Q.
- Topic

# Topics to be Covered

- Topic
- Topic
- Topic
- Topic
- Topic

Question

Nearest Neighbours.

EVERY DAY
IS A CHANGE
TO BE BETTER
alamy

## Recall and Precision

$$TPR = \frac{TP}{\text{Actual } P}$$

$$\frac{TP}{\text{Total Predicted } P}$$

**F1 Score** $= \left\{ 2 \frac{R \times P}{R + P} \right\}$

## Some PYQs

| | |
|---|---|
| Q.12 | Increasing the regularizing coefficient value for a ridge regressor will<br>i) Increase or maintain model bias.<br>ii) Decrease model bias.<br>iii) Increase or maintain model variance.<br>iv) Decrease model variance. |
| (A) | i and iii |
| (B) | i and iv |
| (C) | ii and iii |
| (D) | ii and iv |

λ inc → more generalise
→ Variance ↓↓
→ Bias ↑↑

B

REVISION

## What is Stratified Cross-Validation in Machine Learning?

Cross Validation ⇒ we used this method for finding hyperparameter Ridge Reg, $\lambda$

→ K fold cross validation ⇒ we break K data into K parts.

| 1 | 2 | 3 ... | K |
|---|---|---|---|

→ Train on K-1 parts, Validate on $K^{th}$ part

→ Repeat this step K times.

- In Random distribution of data there can be problem when data has very less portion of any class
- In that case stratified sampling is done in which it is ensured that all the folds/all the groups have data of class 1 & data of class 2 in the same Ratio as that of the Ratio of class 1 and class in whole data.

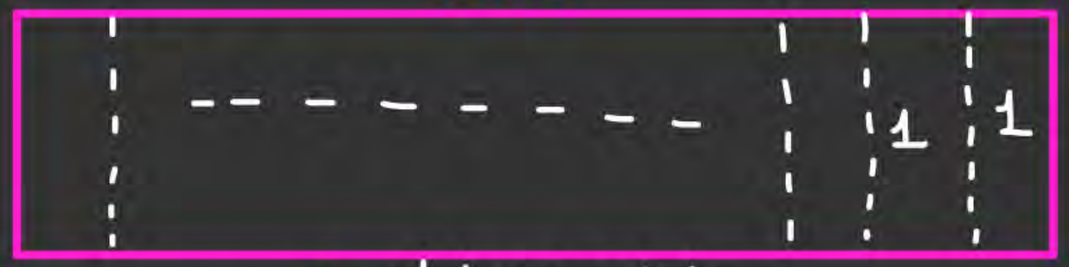
- data
  - → class 1 80
  - → class 2 20

Stratified

| | | | | |
|---|---|---|---|---|
| 20 | 20 | 20 | 20 | 20 |
| 1→16 | 16 | 16 | 16 | 16 |
| 2→4 | 4 | 4 | 4 | 4 |

- LOO-CV

leave one out Cross validation data




→ N data points

→ N folds

and then train on N-1 folds and validate on $N^{th}$ fold

↳ this step is done N times

# Basics of Machine Learning

Bias = Tr. error

Variance = Testing error.

## Some PYQs

Which among the following may help to reduce overfitting demonstrated by a model

i. Change the loss function. → ex Regularisation term in RR

ii. Reduce model complexity. → Yes

iii. Increase the training data. → Yes becoz large dataset lead to understanding pattern of data.

iv. Increase the number of optimization routine steps.

→ we try to Reduce the bias = 0

try to make Cost fxn = 0

→ Complexity inc ⇒ overfit

A. ii and i

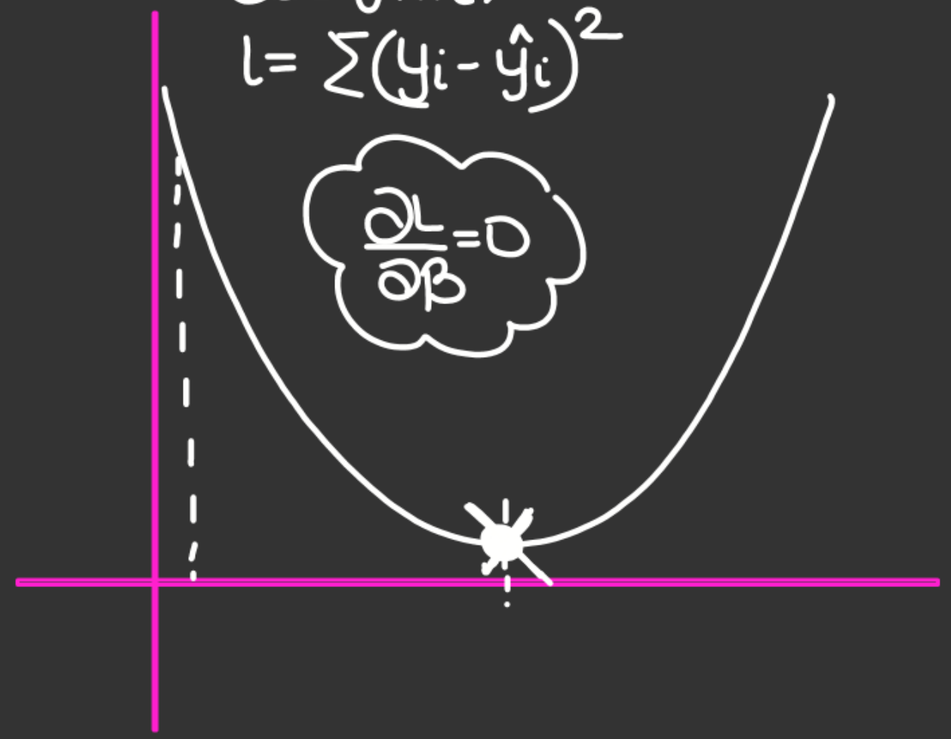
B. ii and iii

C. i, ii, and iii

D. i, ii, iii, and iv

Cost fxn ⇒

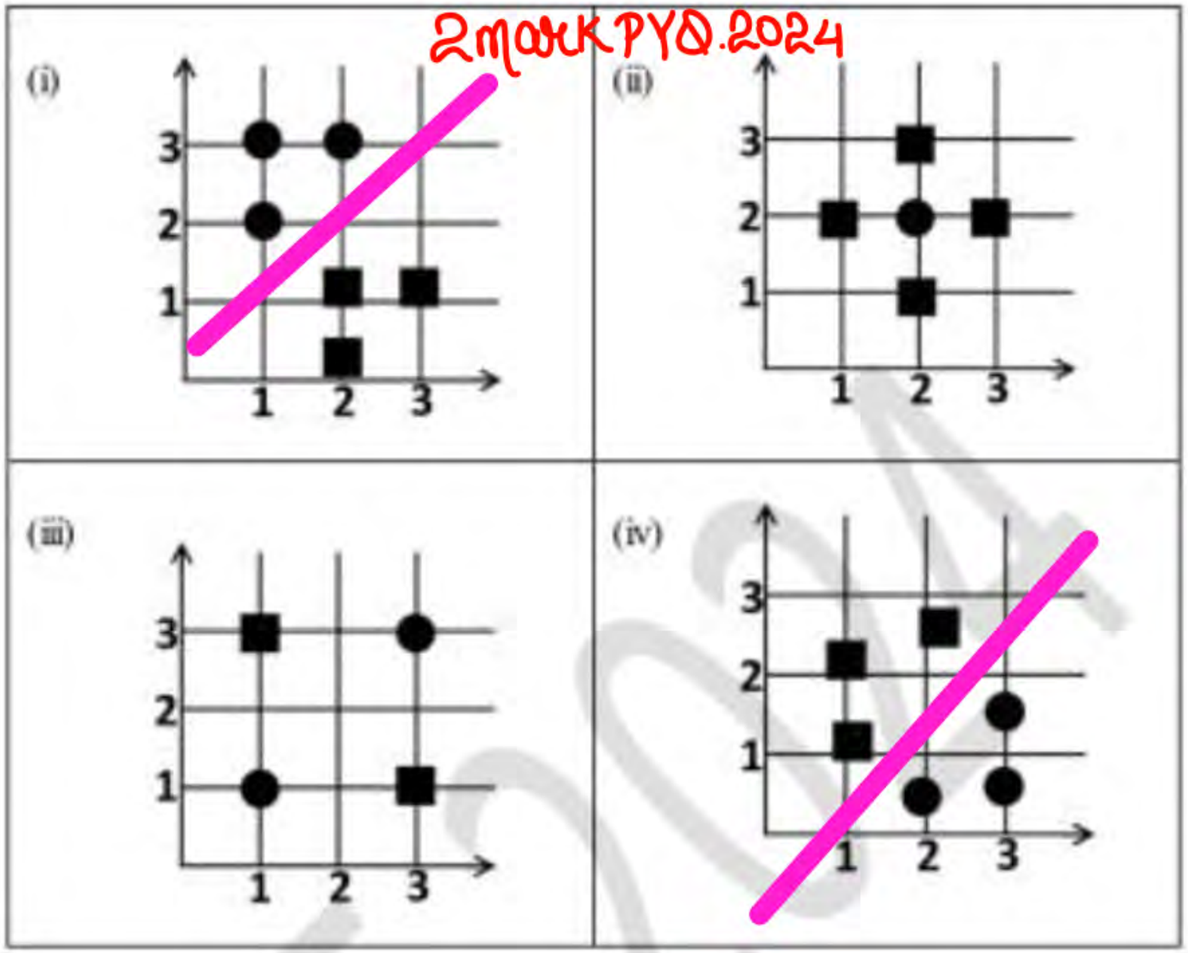
$$l = \sum (y_i - \hat{y}_i)^2$$



# Basics of Machine Learning

Consider the following figures representing datasets consisting of two-dimensional features with two classes denoted by circles and squares.

2MarkPYQ.2024




Which of the following is/are **TRUE**?

| | |
|---|---|
| (A) | (i) is linearly separable. |
| (B) | (ii) is linearly separable. |
| (C) | (iii) is linearly separable. |
| (D) | (iv) is linearly separable. |

# Basics of Machine Learning

You are reviewing four papers submitted to a conference on machine learning for medical expert systems. All the four papers validate their superiority on a standard benchmarking cancer dataset, which has only 5% of positive cancer cases. Which of the experimental settings is acceptable to you?

Case

i) We evaluated the performance of our model through a 5-fold cross-validation process and report an accuracy of 93%.

"2nd algo is best"

2

We computed the average area under the ROC curve through 5-fold cross-validation and found it to be around 0.75 - the highest among all the approaches.

3

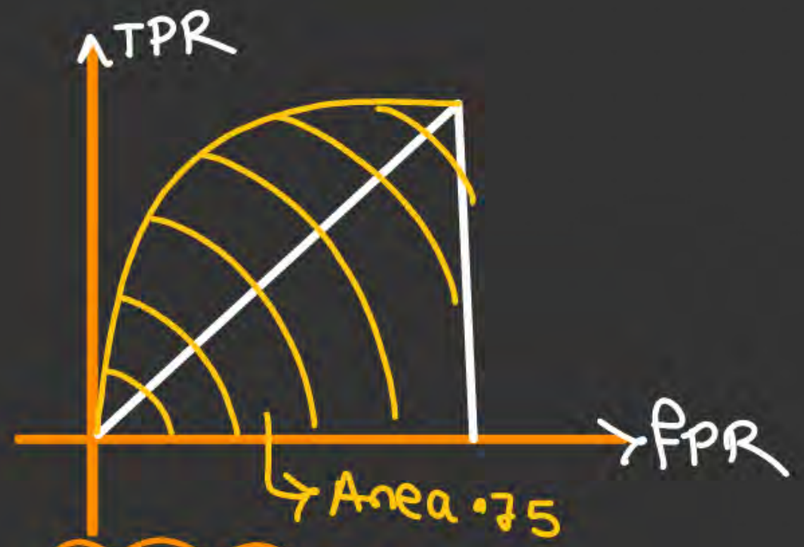
The accuracy on a single left-out test set of our model is 95%, which is the highest among all the different approaches.

5% Pos
95% Neg.

## Case 1

5 folds of data

4 ⇒ Train
1 ⇒ Validation
⇒ 5 times the same steps
⇒ So 5 accuracies on Validation
⇒ 93% ⇒ avg.




TPR > FPR

$$\left(\frac{TP}{\text{Total Actual P}} > \frac{FP}{\text{Total Actual N}}\right)$$

# Basics of Machine Learning

50%

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The accuracy on a single left-out test set of our model is 95%, which is the highest among all the different approaches.

50% P
60% N

3rd Algo best ⇒ 95% accuracy.

# Nearest Neighbour Method

## K-NN method of Supervised Learning

### The general flow of the Supervised Learning Algorithms

linear R
Ridge R
linear C
logistic R

⇒ • Step 1 ⇒ Training from data
⇓
weight parameters
⇓
Testing

**Instance Based Learning**

→ NN algorithm
No Training

→ directly analysis & testing

# Nearest Neighbour Method

## K-NN method of Supervised Learning

### The general flow of the Supervised Learning Algorithms

$(x_1^1, x_1^2)$

$(x_2^1, x_2^2)$

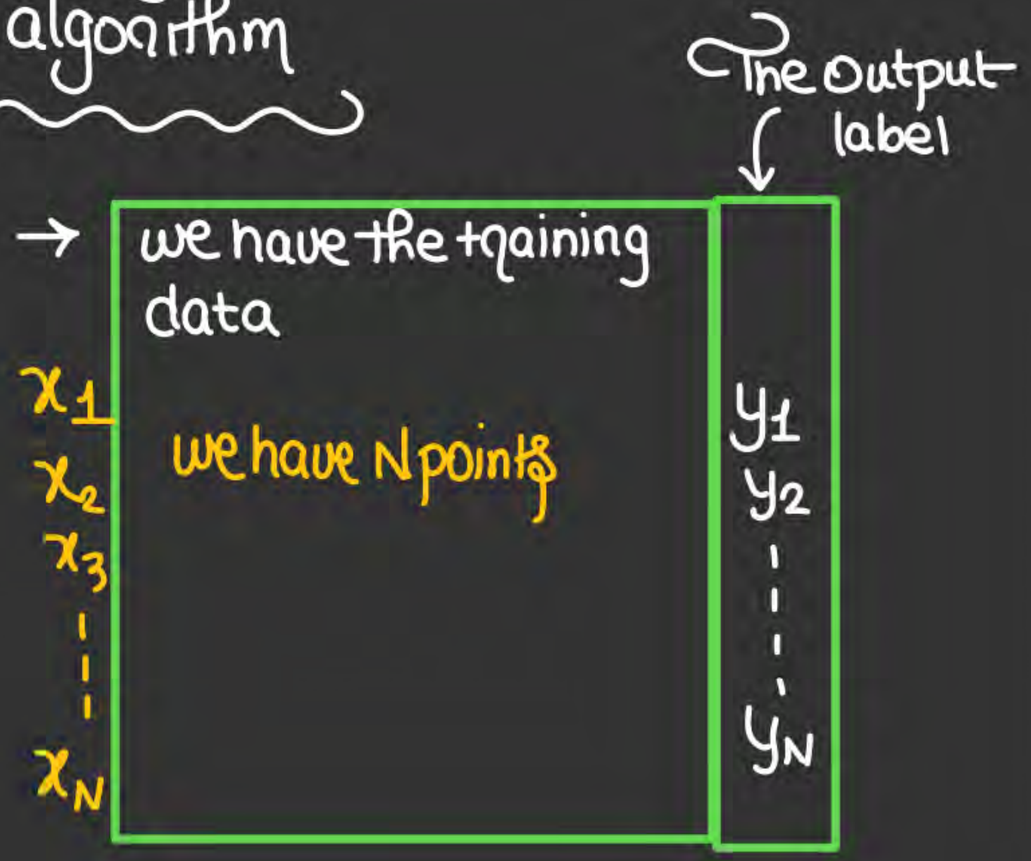
$$\sqrt{(x_1^1 - x_2^1)^2 + (x_1^2 - x_2^2)^2}$$

**Euclidian Distance**

- Euclidian distance b/w 2 points $x_i^0, x_j^0$

$$\sum_{K=1}^{D} \left(x_i^{0K} - x_j^{0K}\right)^2$$

# Basic algorithm




- Whenever we get a new data point for testing i.e to find y label for new data point

- So we have to find the data point similar to the test point in training data

To find the similar points ⇒

- we find euclidian distance of test point from all the training point
- So the 'K' points closest to the test point ⇒ are called similar to test point
- Ypredicted for test point ⇒

Regression

Yp = avg of y of K points

Classification

Yp ⇒ majority class in K points

1. generally K is odd 1, 3, 5...

$K = 1$ → we take only one neighbour
→ NN algorithm

$K = 3, 5, 7, 9 \ldots$ ⇒ K-NN algorithm

# Nearest Neighbour Method

## K-NN method of Supervised Learning

### Lets understand 1-NN with a simplest example...

| Maths | CS | Result |
|---|---|---|
| 4 | 3 | Fail |
| 6 | 7 | Pass |
| 7 | 8 | Pass |
| 5 | 5 | Fail |
| 8 | 8 | Pass |
| 6 | 8 | ?? |

(6, 8) Test

$\sqrt{(6-4)^2+(8-3)^2} = \sqrt{29}$

$\sqrt{(6-6)^2+(8-7)^2} = 1$

$\sqrt{(6-7)^2+(8-8)^2} = 1$

$\sqrt{(6-5)^2+(8-5)^2} \Rightarrow \sqrt{10}$

$\sqrt{(8-6)^2+(8-8)^2} \Rightarrow \sqrt{4}$

• 3NN

$y_P = \max(P, P, P)$

$y_P = P$

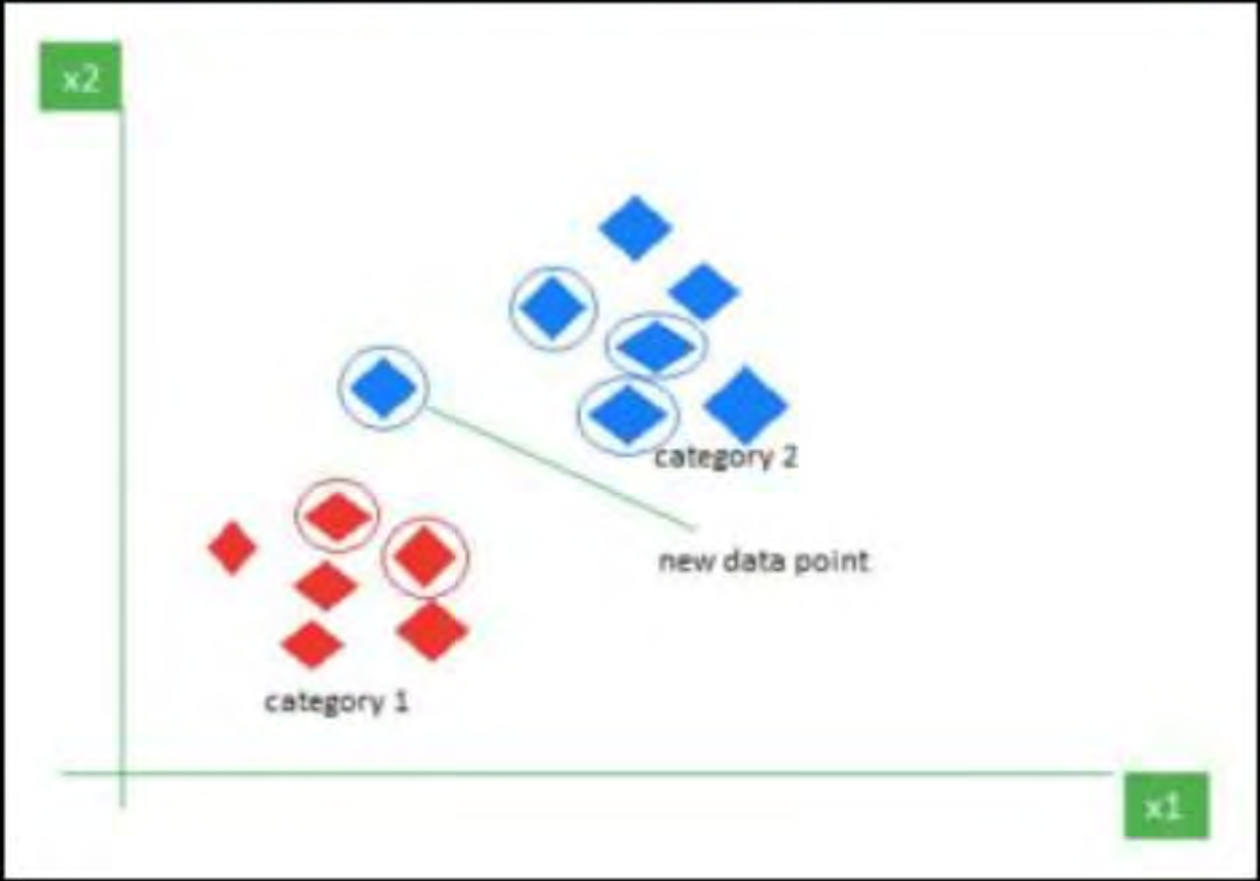
• 1NN ⇒ Pass = $y_P$.

# Nearest Neighbour Method

## K-NN method of Supervised Learning

**Lets understand NN with a simplest example...**

done



# Nearest Neighbour Method

## K-NN method of Supervised Learning

**Lets understand NN with a simplest example...**

done

# Nearest Neighbour Method

## K-NN method of Supervised Learning

## The general flow of the Supervised Learning Algorithms

**Step 1**

Process the data, train the model, find the parameters

**Step 2**

Use the parameters for regression and classification

This is one of the simplest supervised Learning model

This method can be used for
1. Regression
2. Classification

We have instance based learning.

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

**But in Nearest Neighbour the process is as follows...**

**S** → **F** →

- **S**: Similarity Checker.
- **F**: • Calculating the function value

the algo find distance of test point from all training data then we find closest point

$y_p =$ avg / max

This is the lazy learner

This is the instance-based learning

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

### But in Nearest Neighbour the process is as follows...

It is widely disposable in real-life scenarios since it is non-parametric, meaning it does not make any underlying assumptions about the distribution of data

But in KNN

- These are parametric algorithms where we make some assumption on data

- This is non parametric in nature

Because as we do in LR, RR, LC, LR we assume that the data is linearly distributed/ linearly seperable.

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

**But in Nearest Neighbour the process is as follows...**

Similarity measure the distance between the two data points ← euclidian distance.

→ This can work as classifier and regressor...

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

**Similarity between the two points...**

done

Similarity measure the distance between the two data points

Euclidian distance

# Nearest Neighbour Method

1-NN

- $y_p$ = y value of the similar point or closest point.

Step 1 : find the distance of new point from all the other points in the data

Step 2 : Find the distance closest to the point under query

Step 3 : Now Simply assign the value of Y of the closest point to the new point

Finish

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

**Similarity between the two points...**

So we first of all we find the training instance which is closest to the new point

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

**Now how to find the label or value of new point ?**

Now finding the closest or similar instance we then give the same label or value to the new data point

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

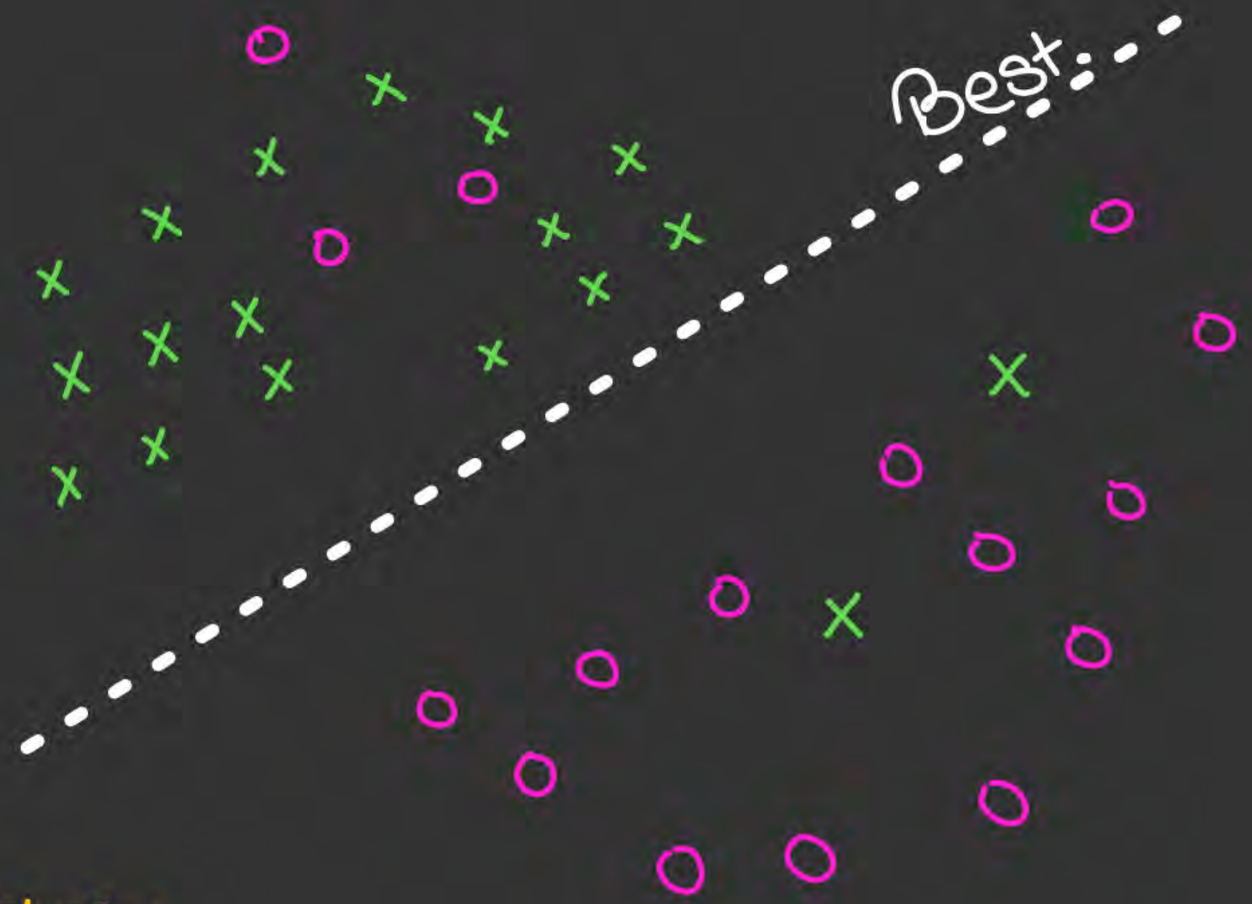
### Decision Boundary in Classification Problem (NN)

What is the problem in NN algorithm ... ?

The motive is to find out the region influenced by each point

What is the Problem in 1-NN algorithm

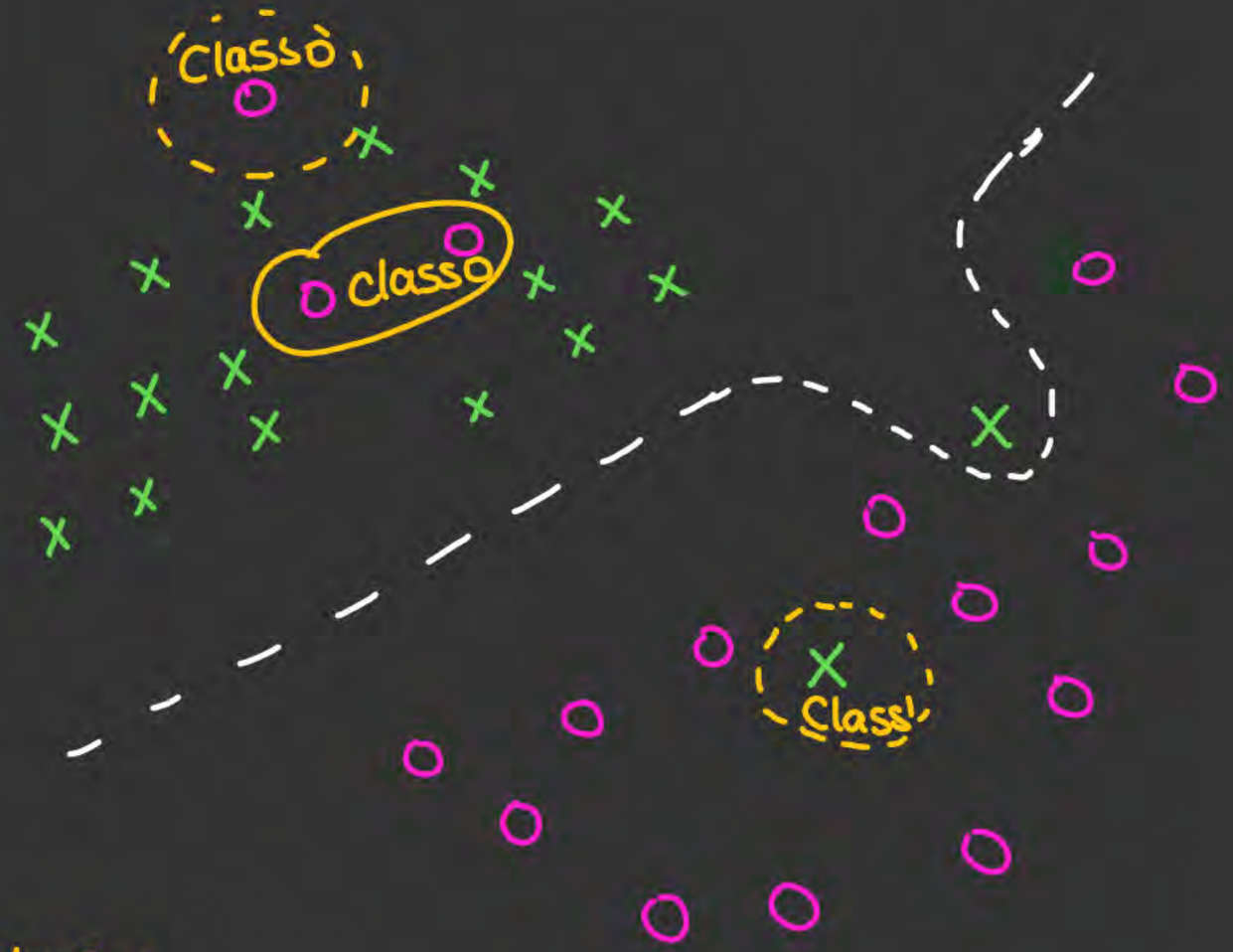
X → class 1
O → class 0




- 1NN → At any location we find the training point, closest to the location and assign the class of closest point.

What is the Problem in 1-NN algorithm

X → class 1
O → class 0

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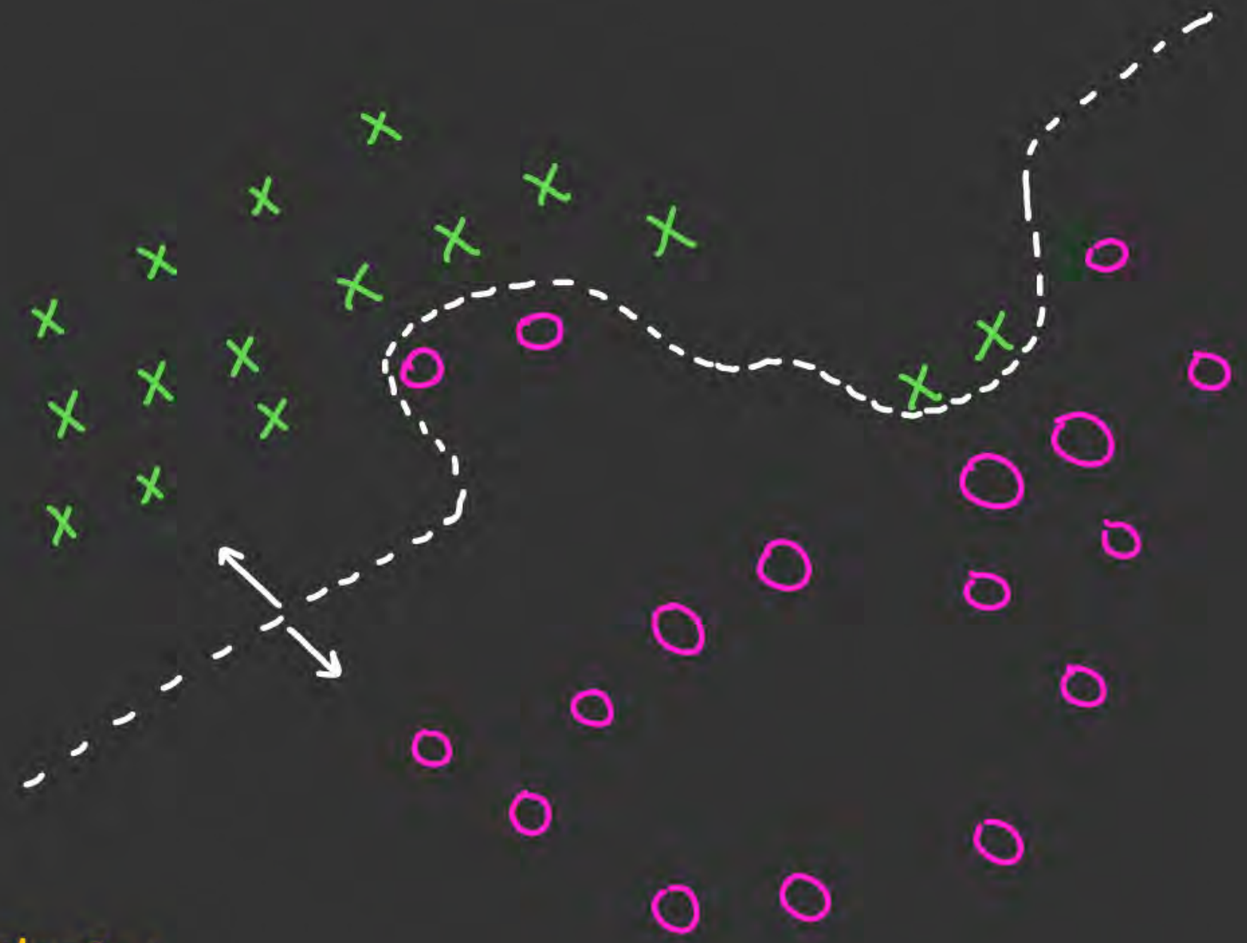


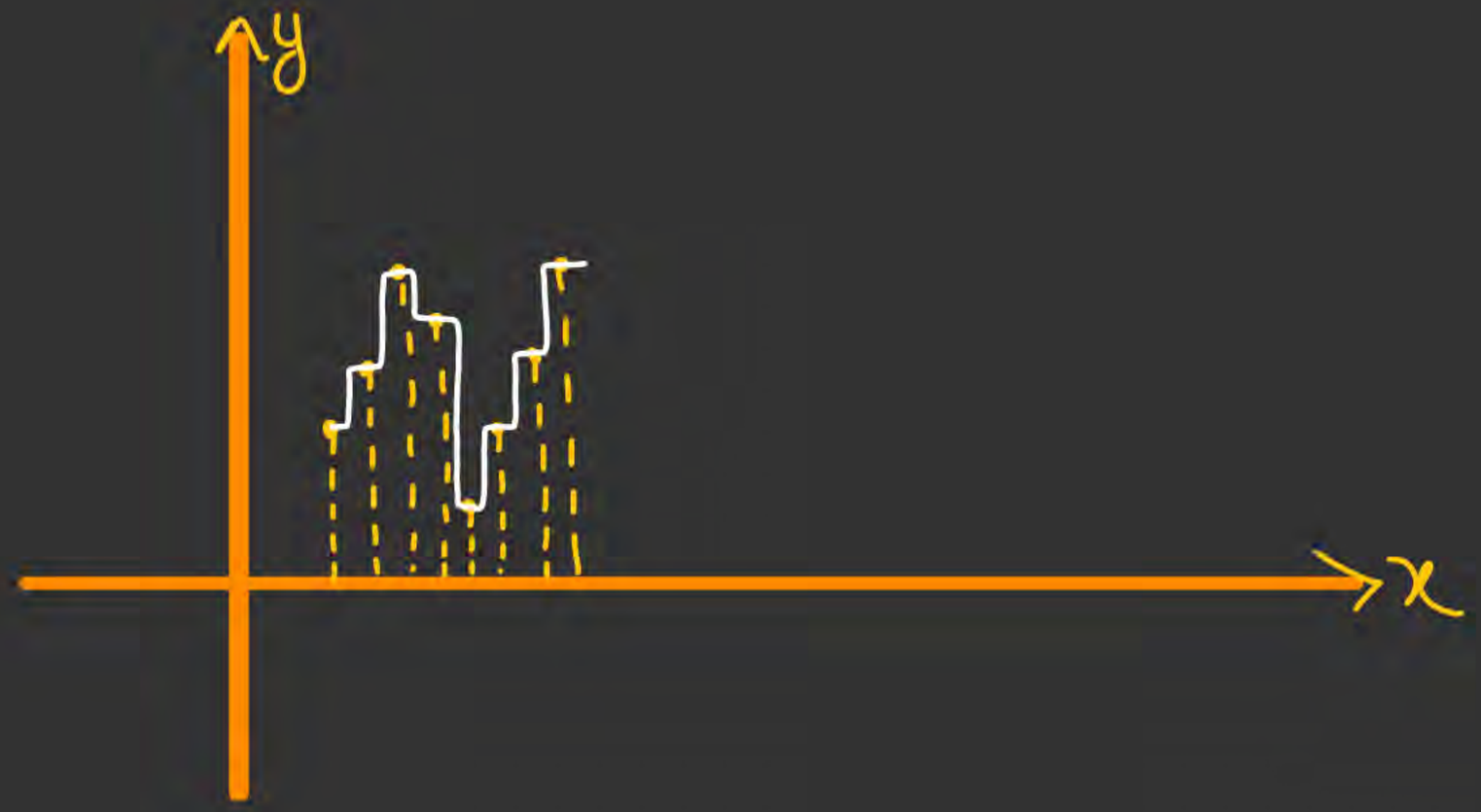
- 1NN ⇒ At any location we find the training point, closest to the location and assign the class of closest point.

Regression

1NN

1D data

| x | y |
|---|---|
| 1 | 3 |
| 2 | 6 |
| 3 | 2 |
| 4 | 7 |


y
x

So 1NN → • highly Complex model

• noise of data included in analysis

• overfitting.

→ Solution K-NN.

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

**Thus we get this decision boundary (Voronoi Diagram)**

So the final decision boundary will be ...

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

The same concept can be used for regression also...

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

### The biggest problem in the NN Method

In NN the whole system is effected by outliers.

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

### How to solve the problem in NN

Move on to K-nearest neighbour technique

Then we do Plurality voting and not majority voting

# Nearest Neighbour Method

## K-NN method of Supervised Learning

**Lets understand KNN with a simplest example...**

| Maths | CS | Result |
|---|---|---|
| 4 | 3 | Fail |
| 6 | 7 | Pass |
| 7 | 8 | Pass |
| 5 | 5 | Fail |
| 8 | 8 | Pass |
| 6 | 8 | ?? |

# What is impact of K on KNN algo.

Dataset D

| $x_1$ | $x_2$ | $y$ |
|---|---|---|
| 1 | 7 | 0 |
| 1.5 | 8 | 0 |
| 2 | 9 | 0 |
| 2.2 | 7 | 0 |
| 1.8 | 10 | 0 |
| 2.3 | 8.5 | 0 |
| 2.9 | 11 | 0 |
| 6 | 7 | 1 |
| 6 | 8 | 1 |
| 6.5 | 6 | 1 |
| 7 | 6 | 1 |
| 7.8 | 7.8 | 1 |

$x \rightarrow y_i = 1$

$o \rightarrow y_i = 0$

More perfect

# K-Nearest Neighbour Method

Step 1 : find the distance of new point from all the other points in the data

Step 2 : Find the distance closest to the point under query

Step 3 : Now Simply assign the average of value of Y of the K closest point to the new point

Finish

# Nearest Neighbour Method

## Nearest Neighbour method of Supervised Learning

### What is K-NN Method of Supervised Learning

K is the hyper parameter

S → F →

# Nearest Neighbour Method

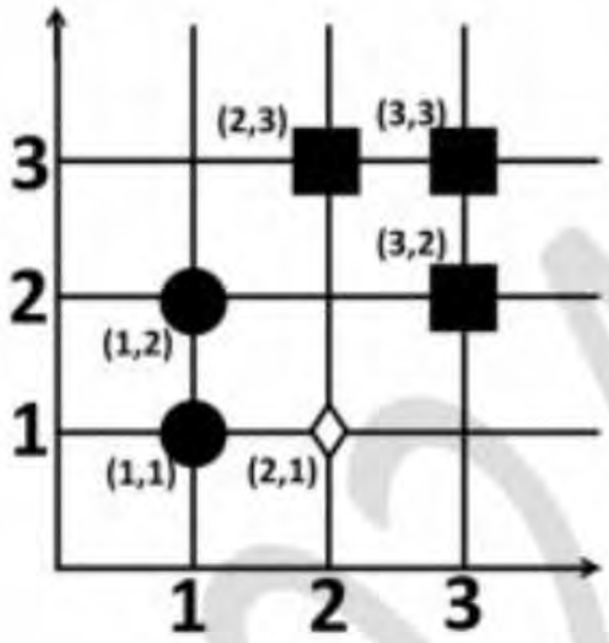
## Nearest Neighbour method of Supervised Learning

### What is K-NN Method of Supervised Learning

K-NN can be used for regression also...

# Nearest Neighbour Method

Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of $k$ in $k$-nearest neighbor algorithm for which the diamond (◇) shaped data point is assigned the label square is _____.



# THANK - YOU